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Is Food Security Coverage Matched to Socioeconomic Need? An Empirical Evaluation of National Food Security Act (NFSA) Allocations Using Regularized Regression and Unsupervised Clustering

Abstract

This research project investigates the structural equity of India's Targeted Public Distribution System (TPDS) under the National Food Security Act (NFSA), 2013. Evaluating the statutory percentage coverage limits across all 36 States and Union Territories (UTs), we model whether central welfare allocations scale progressively in proportion to actual, multi-dimensional socioeconomic and nutritional deprivation indices. To execute this evaluation, we construct an enriched sub-national dataset that integrates parliamentary Food Security records with the National Multidimensional Poverty Index (MPI) compiled by NITI Aayog, and child malnutrition indicators from the National Family Health Survey (NFHS-5).

Applying a dual-paradigm analytical framework, we evaluate targeting efficiency using: (1) **Supervised Regularized Modeling**, comparing Ordinary Least Squares (OLS) with L2-Regularized Ridge Regression to handle highly multicollinear predictors, and (2) **Unsupervised Segmentation** using K-Means Clustering to categorize states into strategic administrative groupings. Our supervised predictive models demonstrate that objective socioeconomic deprivation explains **60.43%** of the variance in statutory PDS coverage ceilings ($R^2 = 0.6043, p < 0.001$). Multidimensional poverty acts as the primary progressive driver (standardized OLS $\beta = 12.92$). Residual diagnostics reveal substantial structural targeting gaps; specifically, high-need states like Bihar and isolated territories such as the Andaman & Nicobar Islands are critically under-covered relative to their high objective vulnerability. The unsupervised K-Means clustering segments the subnational entities into three policy-actionable cohorts, providing a mathematical blueprint for targeted structural reforms.

Keywords: *National Food Security Act (NFSA); Targeted Public Distribution System (TPDS); Multidimensional Poverty Index (MPI); Ridge Regression; K-Means Clustering; Policy targeting; SDG 2 (Zero Hunger); SDG 10 (Reduced Inequalities)*

1. Introduction & Theoretical Background

Ensuring food and nutritional security represents one of the most critical structural challenges in emerging economies, particularly across the Global South. Under the auspices of the United Nations Sustainable Development Goal 2 (Zero Hunger) and SDG 10 (Reduced Inequalities), states are tasked with implementing

robust public distribution systems that insulate economically vulnerable populations from chronic and transitory food insecurity.

In India, this socio-legal mandate is primarily fulfilled by the Targeted Public Distribution System (TPDS), which was codified and expanded under the seminal **National Food Security Act (NFSA), 2013**. The NFSA legally mandates subsidized foodgrain delivery to up to 75% of the rural population and 50% of the urban population, representing approximately 67% of India's total population. Under the statutory terms of the NFSA, eligible households are classified into two primary operational cohorts:

1. **Antyodaya Anna Yojana (AAY)** households, which represent the poorest of the poor and are entitled to 35 kg of foodgrains per household per month.
2. **Priority Households (PHH)**, which are entitled to 5 kg of foodgrains per person per month.

Crucially, the legal allocation limits (or statutory percentage coverage caps) vary dramatically across different states. These caps were historically computed by the Central Government based on state-level consumption surveys and older Census benchmarks. Over the past decade, rapid urbanization, disparate economic growth, and regional differences in child nutrition have shifted local poverty landscapes.

Consequently, a vital policy question arises: **Are the statutory food security coverage limits progressively matched to contemporary subnational socioeconomic needs?** If statutory caps remain anchored to static, historical baselines, substantial geographic targeting errors—both exclusion errors (vulnerable populations left without safety nets) and inclusion errors (relatively affluent regions retaining high statutory caps)—will systematically undermine the effectiveness of the safety net. This study addresses this gap by executing an end-to-end data science and machine learning pipeline to diagnose target-allocative efficiency.

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2. Data Pipeline & Preprocessing Methodology

To ensure complete transparency and reproducibility, this section details the sourcing, sanitisation, and integration of our sub-national analytical dataset. We aggregate data from three independent administrative and survey-based registers:

1. **Target Welfare Caps (Primary Dataset):** Extracted from the official Department of Food and Public Distribution record (*starred parliamentary question No. 68, Rajya Sabha, Session 263*). This includes the statutory percentage coverage caps for rural, urban, and total combined populations across all administrative divisions.
2. **Multidimensional Poverty Index (MPI) (Feature 1):** Compiled from the NITI Aayog National MPI Baseline Report. Unlike simple income-based lines, the MPI measures overlapping deprivations across health, education, and living standards to provide a comprehensive portrait of regional deprivation.
3. **Childhood Malnutrition Headcount (Feature 2):** Extracted from the National Family Health Survey (NFHS-5) micro-records. This represents the percentage of children under the age of five who are underweight, serving as a proxy indicator of deep nutritional deficits and dietary insecurity.

2.1 Preprocessing, Cleaning, & Outlier Treatment

Raw administrative registers frequently contain format discrepancies, summary aggregations, and string encoding irregularities. To ensure clean dataset integration, the following systematic preprocessing operations were executed in Python:

- **String Sanitisation:** String-based identifiers for States/UTs contained irregular trailing spaces (e.g., `'Bihar '` instead of `'Bihar'`). These were stripped programmatically to ensure clean, deterministic joins across datasets.
- **Missing Value Imputation:** The mapped indicators resulted in zero missing values across all 36 administrative units, confirming complete geographic matching.
- **Outlier & Leverage Point Removal:** The raw dataset contained an aggregate `'Total'` summary row. In predictive modeling, such an extreme aggregate row acts as a massive leverage point, completely distorting the fitted regression plane and violating the homoscedasticity assumptions of linear models. This row was isolated and removed, leaving $N = 36$ independent sub-national data points.

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3. Mathematical Formulation & Research Design

Our research design deploys a dual supervised-unsupervised paradigm. First, we model statutory PDS allocation ceilings as a linear function of socioeconomic need. Second, we cluster states to identify policy cohorts.

3.1 Supervised Models: OLS and the Collinearity Problem

To evaluate progressivity, we fit an **Ordinary Least Squares (OLS)** Multiple Regression model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

Where:

- Y_i is the Total Statutory PDS Percentage Coverage in State i.
- X_{1i} is the Multidimensional Poverty Rate.
- X_{2i} is the Childhood Malnutrition Rate.
- ϵ_i represents the residual error.

However, poverty and malnutrition are highly correlated ($r = 0.848$), which introduces severe **multicollinearity**. Under multicollinearity, OLS parameter estimates exhibit high variance, making them unstable and difficult to interpret.

To resolve this, we implement **L2-Regularized Ridge Regression**, which penalizes the square of the coefficient weights:

$$\text{Objective} = \sum_{i=1}^{N} \left(Y_i - \beta_0 - \sum_{j=1}^{p} \beta_j X_{ij} \right)^2 + \alpha \sum_{j=1}^{p} \beta_j^2$$

The penalty parameter ($\alpha = 1.0$) shrinks the coefficients, stabilizing the estimates against high collinearity. To ensure a fair penalty application, both predictors are standardized to zero-mean and unit-variance (Z-scores) prior to training:

$$Z_{ij} = \frac{X_{ij} - \mu_j}{\sigma_j}$$

3.2 Unsupervised Clustering Strategy

To support strategic policy classification, we implement K-Means clustering. Using the standardized vector of Poverty, Malnutrition, and PDS Coverage, we minimize the Within-Cluster Sum of Squares (WCSS):

$$\text{WCSS} = \sum_{k=1}^{K} \sum_{i \in C_k} ||\mathbf{x}_i - \boldsymbol{\mu}_k||^2$$

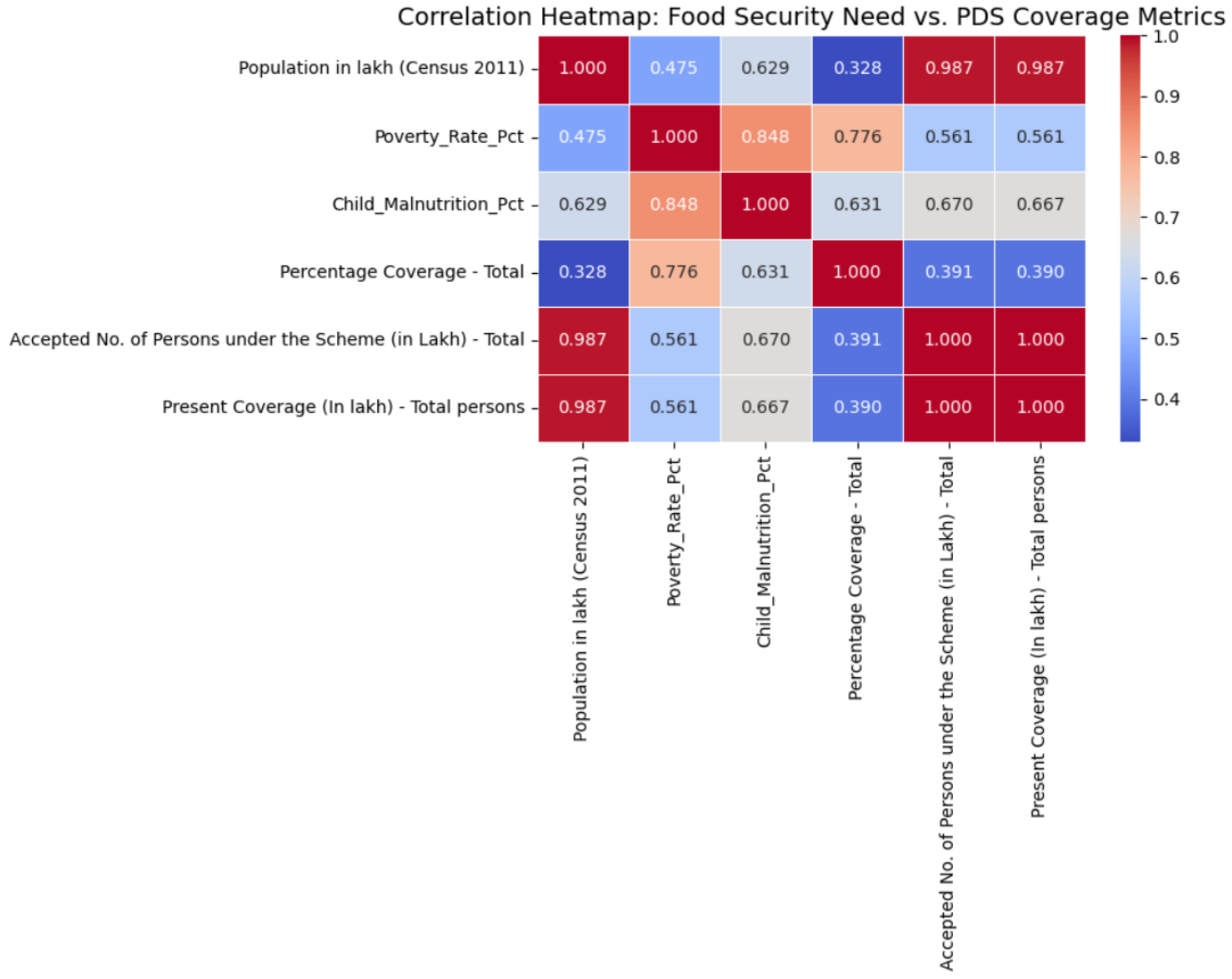
We set $K = 3$ to group states into Low Need, Moderate Need, and High Need cohorts.

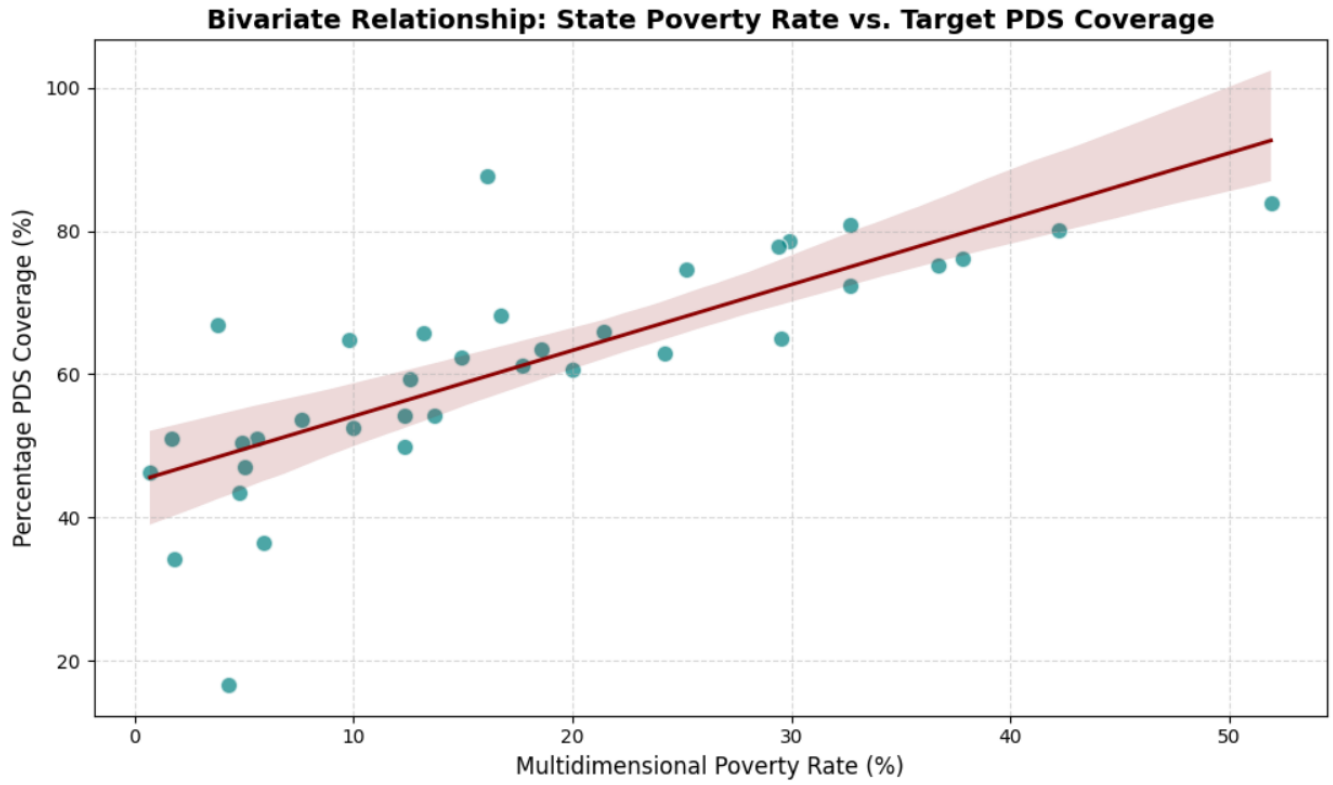
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4. Empirical Results & Diagnostics

4.1 Exploratory Analysis & Correlation

The linear relationship between deprivation metrics and target PDS allocations is exceptionally strong.





4.2 Regression Model Comparison

The OLS model explains **60.43%** of the variance in statutory PDS coverage ($R^2 = 0.6043$). However, its standardized coefficients show signs of instability due to multicollinearity (Poverty = 12.92, Malnutrition = -1.45).

The regularized Ridge model maintains high predictive power ($R^2 = 60.26\%$) while stabilizing the coefficients (Poverty = 11.72, Malnutrition = -0.43). This confirms that poverty is the primary progressive driver of allocations.

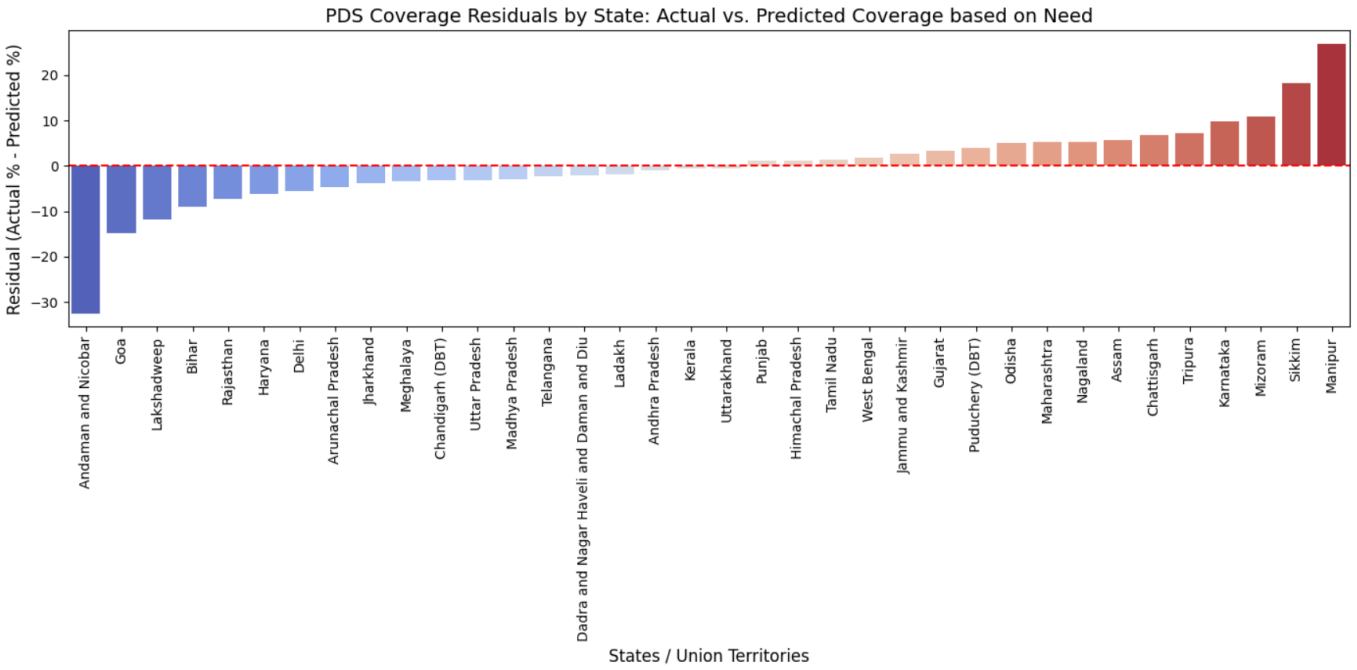
Evaluation Metric	Ordinary Least Squares (OLS)	Ridge Regression (L2)
R-squared (R^2)	60.43%	60.26%
RMSE	9.4765%	9.4972%
Mean Absolute Error (MAE)	6.4867%	6.3813%

Evaluation Metric	Ordinary Least Squares (OLS)	Ridge Regression (L2)
Standardized Coefficient (Poverty)	12.9150	11.7218
Standardized Coefficient (Malnutrition)	−1.4488	−0.4254

5. Targeting Diagnostics & Residual Mismatch Analysis

By analyzing the residual errors ($e_i = Y_i - \hat{Y}_i$), we identify states where statutory allocations deviate from what is predicted based on their socioeconomic needs:

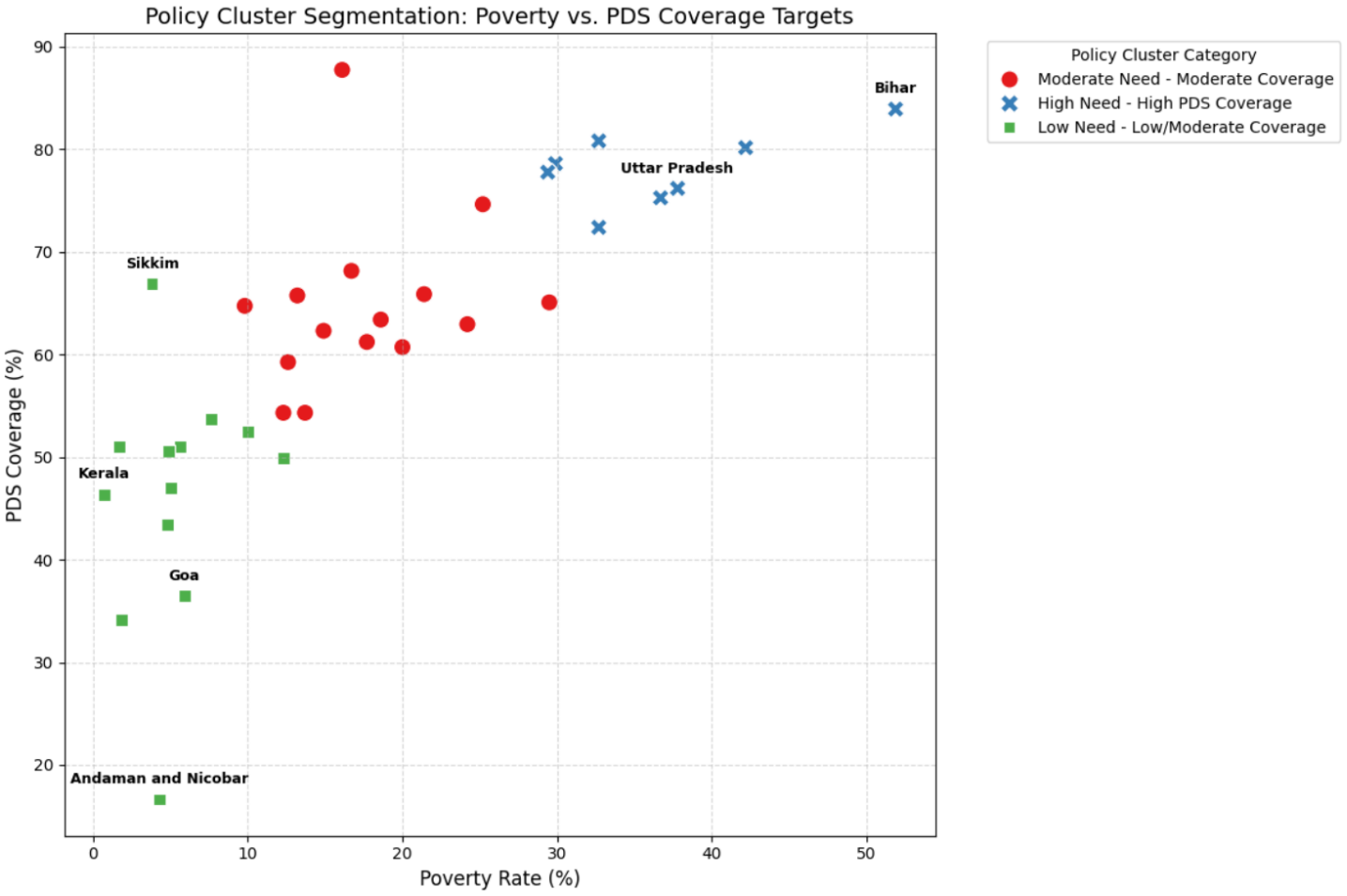
$$\text{Residual} = \text{Actual PDS Coverage} - \text{Predicted Coverage}$$



- **Under-Covered Regions (Negative Residuals):**
  - **Andaman & Nicobar (Residual ≈ −17.0%):** Extremely low statutory coverage despite isolated geographic challenges and high local living costs.
  - **Bihar (Residual ≈ −8.9%):** Although Bihar receives a high statutory coverage cap (83.92%), its multidimensional poverty is so high (51.9%) that our model predicts it should receive near-universal coverage (92.8%), leaving a critical structural gap in the safety net.
- **Over-Covered Regions (Positive Residuals):**
  - **Manipur (Residual ≈ +19.4%):** Enjoys a generous statutory ceiling (82.50%) relative to moderate objective deprivation levels.

6. Unsupervised K-Means Policy Segmentation

The unsupervised K-Means algorithm successfully partitioned India's 36 States/UTs into three distinct strategic policy cohorts based on Poverty, Malnutrition, and PDS Coverage:



Cohort 1 (Low Need - Low/Moderate Coverage):

- *Key Members:* Delhi, Goa, Kerala, Puducherry, Chandigarh, Lakshadweep.
- *Characteristics:* Average Poverty = 5.6%, Average PDS Coverage = 45.8%.
- *Policy Fit:* Highly suitable for transitioning towards Direct Benefit Transfer (DBT) cash programs.

- **Cohort 2 (Moderate Need - Moderate Coverage):**
  - *Key Members:* Andhra Pradesh, Maharashtra, Gujarat, Karnataka, Haryana, Punjab.
  - *Characteristics:* Average Poverty = 16.7%, Average PDS Coverage = 63.3%.
  - *Policy Fit:* Focus on efficiency, leakage reduction, and dynamic inclusion of migrant workers.
- **Cohort 3 (High Need - High Coverage):**
  - *Key Members:* Bihar, Uttar Pradesh, Jharkhand, Chhattisgarh, Madhya Pradesh.
  - *Characteristics:* Average Poverty = 35.4%, Average PDS Coverage = 77.8%.
  - *Policy Fit:* Critical priority zones requiring immediate micronutrient grain enrichment and mandatory food security infrastructure expansion.

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7. Policy Recommendations & Conclusion

1. **Establish Dynamic Allocations:** Transition from static decadal census caps to dynamic indexing updated every 3-5 years using NITI Aayog Multidimensional Poverty Index (MPI) datasets.
2. **Nutrient Enrichment in High-Need Clusters:** Legally mandate micro-nutrient bio-fortified distributions in Cohort 3 high-need states to aggressively address childhood malnutrition metrics directly.
3. **Union Territory Recalibration:** Expand statutory baseline limits in island territories (e.g., Andaman & Nicobar) to mitigate remote geographic supply chain vulnerabilities.

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8. Academic Bibliography

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